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Chapter 2: Probability

Hoang Van Ha
University of Science, VNU - HCM
hvha@hcmus.edu.vn

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Random events

Random experiment (*Random experiment*)

It is the realization of a certain set of conditions (specific experiment or observation of a certain phenomenon), which can be repeated many times. The outcome of the experiment cannot be determined in advance.

Example 1

Random experiment	Outcome
Tossing a coin	*Heads, tails*
Rolling a die	{⚀, ⚁, ⚂, ⚃, ⚄, ⚅}
Randomly answering a multiple-choice question	*Correct/Incorrect*

Random Events

- The set of all possible outcomes when performing an experiment is called the sample space or space of elementary events (*sample space*), denoted by Ω.
- Each outcome of a random experiment ω ($\omega \in \Omega$) is called a simple event (*simple event*).
- A subset of the sample space is called a random event (*event*). Denoted by $A, B, C, \ldots$
- The event that always occurs when performing the experiment is the certain event, denoted by Ω.
- The event that never occurs is called the impossible event (or empty event) (*empty event*), denoted by $\emptyset$.

Random event

Example 2

Roll a die once. Let $\omega_i =$ "the upper face has i dots $= i$". The space of elementary events

$$\Omega = \{\omega_1, \omega_2, \ldots, \omega_6\} = \{\text{⚀}, \text{⚁}, \text{⚂}, \text{⚃}, \text{⚄}, \text{⚅}\}$$

$A = \{\text{⚀}, \text{⚂}, \text{⚄}, \} =$ "odd dots" $\searrow$

$B = \{\text{⚁}, \text{⚃}, \text{⚅}\} =$ "even dots" $\rightarrow$ Random event

$C = \{\text{⚄}, \text{⚅}\} \quad =$ "dots > 4" $\nearrow$

Relations between events

Implication

A implies B, denoted $A \subset B$, if A occurs then B occurs. We also say A is an event favorable to B.

Example 3

Roll a die.

Let A_i *be the event of getting* i *dots* $\left(i = \overline{1,6}\right)$,

B *be the event that the number of dots is divisible by* 3,

C = *"Even number of dots"* ,

P_2 = *"Even prime number of dots"*,

Then we have $A_2 \subset C, A_3 \subset B, A_2 \subset P_2, P_2 \subset A_2$.

Relations between events

Equivalence

A is equivalent to B, denoted $A = B$, if A occurs then B occurs and vice versa.

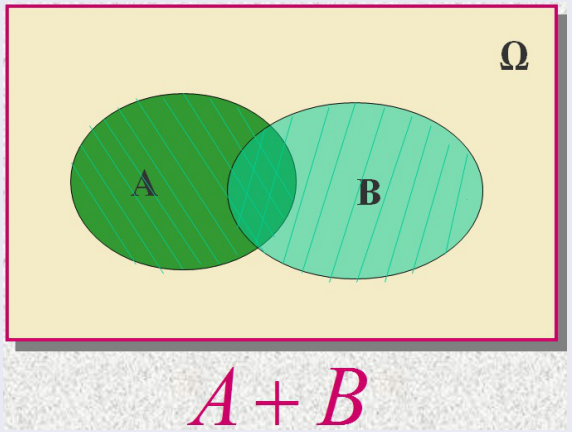
Example 4

In Example 3: $A_2 = P_2$.

Operations on events

Union event (*union*)

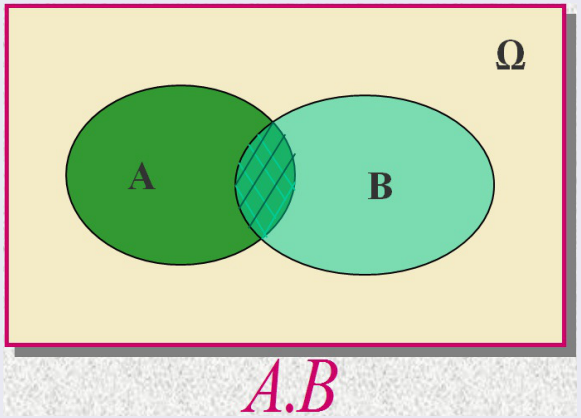
The union of A and B, denoted by $A + B$ or $A \cup B$, is the event that occurs if A or B occurs (at least one of the two events occurs).



Operations on events

Intersection

The intersection of A and B, denoted by $A.B$ or $A \cap B$, is the event that occurs if A and B occur simultaneously.



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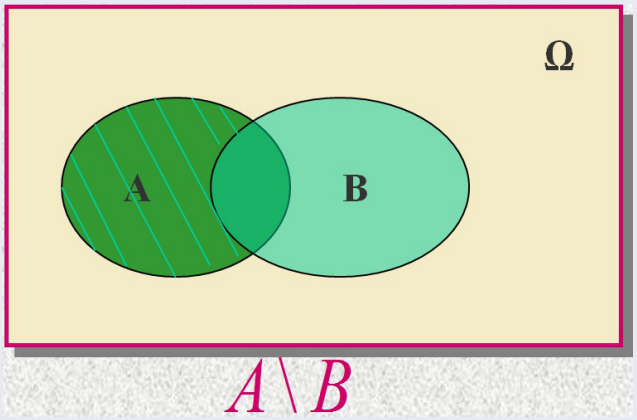
Difference of events

The difference of events A and B, denoted by $A \setminus B$, is the event that occurs if A occurs but B does not occur.



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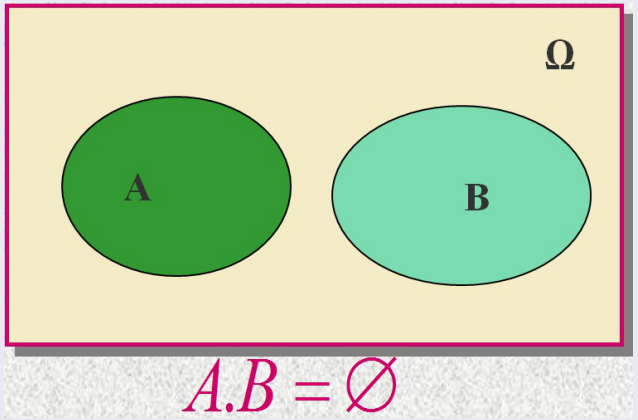
Mutually exclusive events (*mutually exclusive*)

A is mutually exclusive with B if A and B cannot occur simultaneously, denoted by $A \cap B = \emptyset$.$\hat{}$

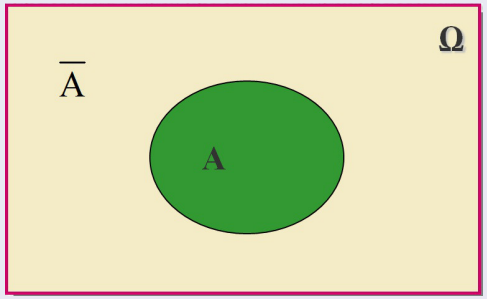


A sequence of events $A_1, A_2, \ldots, A_n$ is called pairwise mutually exclusive if $A_i \cap A_j = \emptyset, \forall i \neq j$.

Operations on events

Complementary event (complement)

The complementary event of A, denoted by $\overline{A}$, is the event that occurs when A does not occur and vice versa, meaning $\begin{cases} A \cup \overline{A} = \Omega \\ A \cap \overline{A} = \emptyset \end{cases}$ or $\overline{A} = \Omega \setminus A$.



Properties

$$\overline{A \cup B} = \overline{A} \cap \overline{B}$$

$$\overline{A \cap B} = \overline{A} \cup \overline{B}$$

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Example 5

A machine has 3 components operating independently. The machine stops functioning when all 3 components are defective. Define the events:

$$A_i\text{: "The } i\text{-th component is defective", } i = 1, 2, 3$$

Express the following events in terms of A_i:

(a) B = *"At most two components are defective"*

(b) C = *"At least one component is defective"*

(c) D = *"At least two components are defective"*

(d) E = *"The machine stops functioning"*

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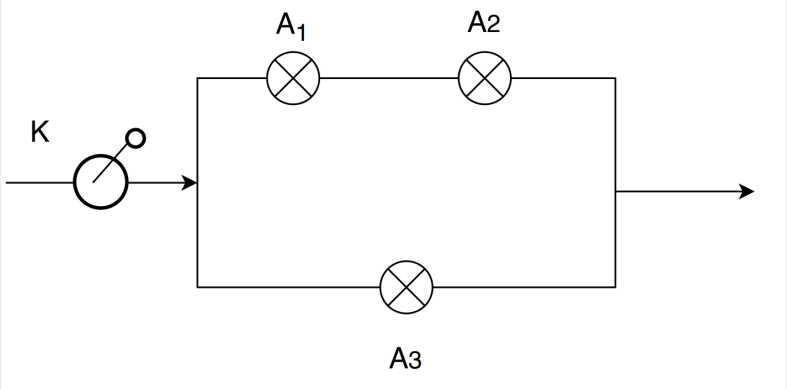
Example 6

Consider an electrical circuit diagram as shown in the figure, consisting of a switch K and light bulbs A_1, A_2, and A_3. The network loses power (B) only when the light bulbs fail or switch K fails. Express B in terms of $A_i, i = \overline{1,3}$ and K.



Concept and Definitions of Probability

Concept of Probability

The probability of an event A is a number characterizing the likelihood of the occurrence of event A in a corresponding experiment. It is denoted as $P(A)$.

Remark 1

- *The larger $P(A)$ is (closer to 1), the more likely A is to occur.*
- *The smaller $P(A)$ is (closer to 0), the less likely A is to occur.*

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Definition 1.1 (Classical definition of probability)

If in a trial there are total n equally likely elementary events, meaning $P(\omega_1) = P(\omega_2) = \ldots = P(\omega_n) = \frac{1}{n}$, among which there are m outcomes favorable to event A, then the probability of A, denoted by $P(A)$, is the ratio $\frac{m}{n}$.

$$P(A) = \frac{card(A)}{card(\Omega)} = \frac{m}{n} = \frac{\textit{Number of outcomes favorable to } A}{\textit{Total number of possible outcomes}}. \quad (1)$$

Alternative notation: $card(A) = |A|$.

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Example 7

A box contains 3 *white balls and* 5 *red balls identical in size.* 3 *balls are drawn at random from the box. Calculate the probability of obtaining*

(a) 3 *red balls.*

(b) 2 *white balls and* 1 *red ball.*

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Advantages and disadvantages

- Advantage: The exact value of probability can be calculated without conducting the experiment.
- Disadvantage: Because it requires a finite number of outcomes and their equiprobability, whereas in reality, many experiments do not possess these properties. Therefore, it is necessary to introduce other definitions of probability to overcome the above limitations.

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Definition 1.2 (Statistical Definition of Probability)

Perform n trials. Suppose event A occurs m times. Then m is called the frequency *of occurrence of event A in n trials, and the ratio $\frac{m}{n}$ is called the* relative frequency *of occurrence of event A in n trials, denoted by $f_n(A) = \frac{m}{n}$. If the trials are performed infinitely many times ($n \to \infty$), the relative frequency of event A approaches a definite number called the probability of event A.*

$$P(A) = \lim_{n\to\infty} f_n(A) = \lim_{n\to\infty} \frac{m}{n}. \tag{2}$$

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Example 8

To investigate the likelihood of obtaining a head when tossing a coin, the coin was tossed many times and the following results were obtained:

Experimenter	*Number of tosses n*	*Number of heads m*	*Relative frequency* $\left(\frac{m}{n}\right)$
Buffon	*4040*	*2048*	*0.5069*
Pearson	*12000*	*6019*	*0.5016*
Pearson	*24000*	*12012*	*0.5005*

The table above shows that as the number of tosses increases, the relative frequency of heads appearing $\frac{m}{n}$ *approaches* $\frac{1}{2}$.

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Advantages and disadvantages

- Advantage: does not require the trial to have a finite number of equally likely events; probability is calculated based on empirical observations, so it is widely applied.
- Disadvantage: requires repeating the trial many times. In many practical problems, this is not feasible due to conditions and the cost of conducting trials...

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Definition 1.3 (Geometric Definition)

Consider an equiprobable experiment where the sample space has infinite elements and is represented as a geometric region Ω with a defined measure (length, area, volume). Event $A \subset \Omega$ is represented by the geometric region A. Then, the probability of A occurring is defined by:

$$P(A) = \frac{\textit{Measure of region } A}{\textit{Measure of region } \Omega}. \tag{3}$$

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Example 9 (The meeting problem)

Two people agree to meet at a location between 11 *and* 12 *o'clock. They agree that the person who arrives first will wait for only* 20 *minutes, and leave if they do not meet. Assume that each person's arrival is random. Find the probability that the two people meet.*

Example 10

A metal rod of length l (m) is broken into 3 pieces at random. Calculate the probability that those 3 pieces form a triangle.

Concepts and definitions of probability

Properties of probability

1. $0 \leq P(A) \leq 1$.
2. $P(\emptyset) = 0$ and $P(\Omega) = 1$.
3. If $A \subset B$ then $P(A) \leq P(B)$.
4. $P(\overline{A}) = 1 - P(A)$.

Addition rule of probability

1. For arbitrary events:
   1. For any A, B, we have
   $$P(A+B) = P(A) + P(B) - P(A.B). \tag{4}$$
   2. $A_1, A_2, \ldots, A_n$:
   $$P\left(\sum_{i=1}^{n} A_i\right) = \sum_{i=1}^{n} P(A_i) - \sum_{1 \le i < j \le n} P(A_i A_j) + \cdots + \\ + (-1)^{n-1} P(A_1.A_2 \ldots A_n). \tag{5}$$

Probability Addition Rule

2. For mutually exclusive events
   1. If A, B are mutually exclusive, we have $P(A+B) = P(A) + P(B)$.
   2. $A_1, A_2, \ldots, A_n$ are pairwise mutually exclusive $\left(A_i.A_j = \emptyset, \forall i \neq j\right)$,

$$P\left(\sum_{i=1}^{n} A_i\right) = \sum_{i=1}^{n} P(A_i),$$

Example 11

The proportion of people with heart disease in a residential area is 9%, with hypertension is 12% and with both diseases is 7%. Randomly select one person. Calculate the probability that the person

(a) *Has heart disease or hypertension.*

(b) *Has neither of the above diseases.*

Conditional probability formula

Definition 2.1 (Conditional probability)

Let A and B be two events with $P(B) > 0$. The probability of event A occurring given that event B has occurred is

$$P(A|B) = \frac{P(AB)}{P(B)}, \quad P(B) > 0.. \tag{6}$$

Similarly, with $P(A) > 0$, the probability of event B occurring given that event A has occurred is

$$P(B|A) = \frac{P(AB)}{P(A)}, \quad P(A) > 0. \tag{7}$$

Conditional probability formula

Properties of conditional probability

- $0 \leq P(A|B) \leq 1$.
- $P(B|B) = 1$.
- If $AC = \emptyset$ then $P[(A + C)|B] = P(A|B) + P(C|B)$.
- $P(\bar{A}|B) = 1 - P(A|B)$.

Conditional probability formula

Example 12

A standard deck of 52 playing cards is well-shuffled. One card is chosen at random. Given that a red card is chosen, calculate the probability that it is the Ace of Hearts.

Example 13

A group consists of 300 people, including 200 men and 100 women. Among the 200 men, 100 are smokers. Among the 100 women, 20 are smokers. A person is chosen at random

(a) *Given that a woman is chosen, calculate the probability that she is a smoker?*

(b) *Given that a smoker is chosen, calculate the probability that the person is male?*

Conditional probability formula

Example 14

The table below summarizes the analysis results of 150 galvanized steel samples for coating weight and surface roughness:

		Coating weight	
		High	Low
Surface roughness	High	12	16
	Low	88	34

If a steel sample is selected at random, given that the selected sample has a high coating weight, what is the probability that the sample has high surface roughness?

Multiplication rule of probability

Corollary 2.1 (Multiplication rule)

For arbitrary events A and B, we have

$$P(AB) = P(A|B)P(B) = P(B|A)P(A).$$

General multiplication rule

Let A_i $(i = 1, ..., n)$ be a family of n events, then

$$P(A_1A_2...A_n) = P(A_1)P(A_2|A_1)P(A_3|A_1A_2)...P(A_n|A_1A_2...A_{n-1}).$$

Multiplication Rule of Probability

Example 15

Suppose that your ATM card password is a sequence of 6 digits and you have forgotten the last digit of this password. You select this last digit randomly. It is known that if you enter the wrong password more than 3 times, your ATM card will be locked. Calculate the probability that you enter the correct password without trying more than 3 times. If it is known that the last digit is an odd number, what is this probability?

Example 16

This semester, a student is allowed to take the Probability Theory and Mathematical Statistics exam 3 times. The probability that the student passes on the first attempt is 0.5*. If the student fails the first time, the probability of passing on the second attempt is* 0.7*. If the student fails both the first two times, the probability of passing on the third attempt is* 0.9*. Calculate the probability that this student passes this semester.*

Independence of events

Two independent events

Two events A and B are said to be independent if

$$P(AB) = P(A).P(B). \tag{8}$$

It follows that if A is independent of B, then

$$P(A|B) = P(A),$$
$$P(B|A) = P(B).$$

Independence of events

Example 17

Investigate whether the genders of children in families with 2 children (in birth order) are independent?
The space of elementary events of the experiment: $\Omega = \{TT, TG, GT, GG\}$
Define:
$A =$ *"The first child is a boy"* $= \{TT, TG\}$
$B =$ *"The second child is a girl"* $= \{TG, GG\}$
We have:

$$P(A) = \frac{2}{4} = \frac{1}{2} \quad \text{and} \quad P(B) = \frac{2}{4} = \frac{1}{2}$$

and $P(AB) = \frac{1}{4} = P(A).P(B)$. *Thus A, B are independent.*

Independence of Events

n Independent Events

Events $A_1, A_2, ..., A_n$ are said to be independent if they satisfy:

$$P(A_iA_j) = P(A_i)P(A_j),$$
$$P(A_iA_jA_k) = P(A_i)P(A_j)P(A_k),$$
$$P(A_1A_2...A_n) = P(A_1)P(A_2)...P(A_n),$$

for every combination of 2 indices (i, j), 3 indices (i, j, k), ..., up to n indices.

Independence of Events

Example 18

Consider a random experiment with equally likely outcomes:

ω	ω_1	ω_2	ω_3	ω_4
$\mathbb{P}(\omega)$	$\frac{1}{4}$	$\frac{1}{4}$	$\frac{1}{4}$	$\frac{1}{4}$

Let: $A = \{\omega_1, \omega_4\}$, $B = \{\omega_2, \omega_4\}$, $C = \{\omega_3, \omega_4\}$ *then:*

$$P(AB) = P(A).P(B),$$
$$P(AC) = P(A).P(C),$$
$$P(BC) = P(B).P(C),$$

but:

$$P(ABC) \neq P(A).P(B).P(C).$$

Note:

Pairwise independence does not imply mutual (total) independence.

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Example 19

A box contains 10 balls numbered from 1 to 10. A ball is chosen at random. Let R be the event that an even-numbered ball is chosen, S be the event that the ball number is ≤ 6, and T be the event that the ball number is ≤ 4. Examine the independence of the pairs of events (R, S), (R, T), and (S, T).

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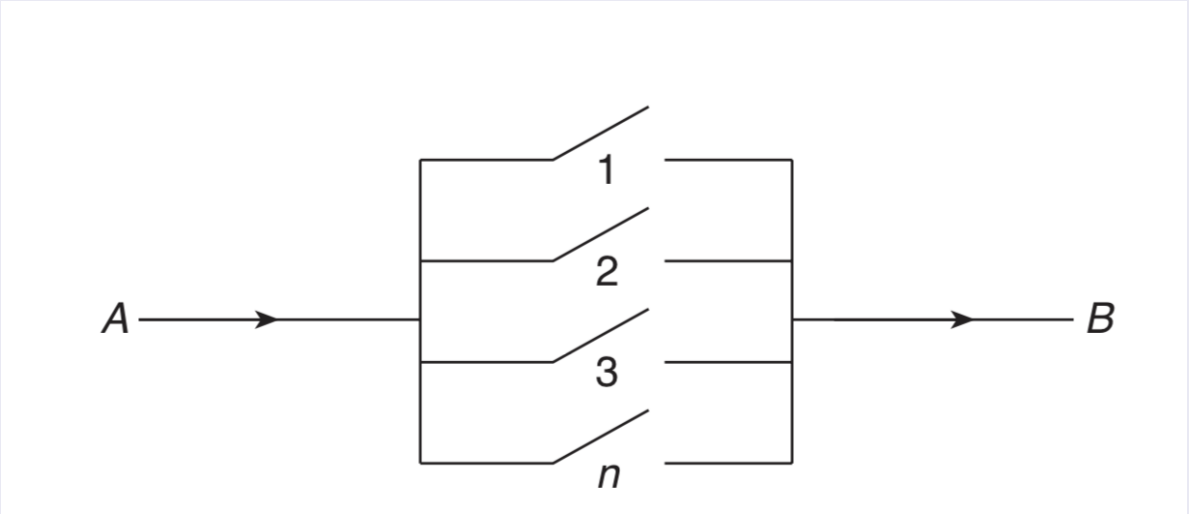
Proposition 2.2

If event A is independent of B, then A is also independent of $\bar{B}$.

Independence of Events

Example 20

A system of n components is called a parallel system if it functions when at least one of the components functions. Suppose the components function independently, and the probability that the i-th component functions is p_i, $i = 1, \ldots, n$. Calculate the probability that the system functions.

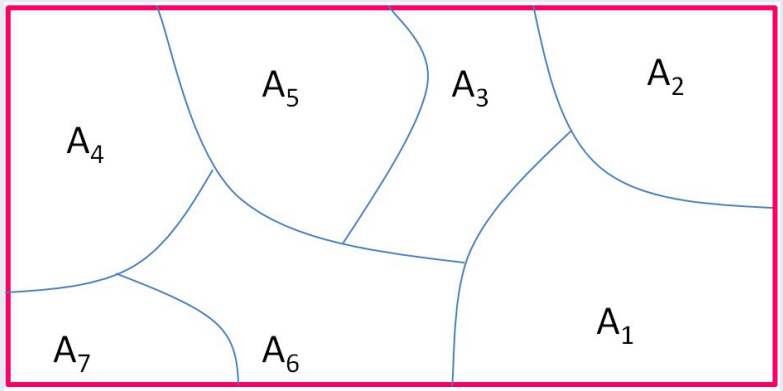


Exhaustive System of Events

Exhaustive System of Events

A sequence of n events $A_1, A_2, \ldots, A_n$ is called an exhaustive system of events (or a partition) if:

$$\begin{cases} A_i \cap A_j = \emptyset, \forall i \neq j, i, j = \overline{1, n} \\ A_1 \cup A_2 \cup \cdots \cup A_n = \Omega. \end{cases}$$



$\{A_i\}_{i=1}^n$ is also called a partition of Ω.

Total Probability Rule

Definition 2.2 (Total Probability Rule)

Let A_i $(i = 1, ..., n)$ be an exhaustive system of events and B be an event related to this system, then:

$$\begin{aligned} P(B) &= P(A_1)P(B|A_1) + P(A_2)P(B|A_2) + ... + P(A_n)P(B|A_n) \\ &= \sum_{i=1}^{n} P(A_i)P(B|A_i). \end{aligned} \tag{9}$$

Total Probability Rule

Example 21

In a certain population, the number of men is half the number of women. The probability that a man has heart disease is 0.06 *and for a woman is* 0.036*. A person is chosen at random; calculate the probability that this person has heart disease.*

Example 22

A person has 3 different email accounts. Most of their messages, 70%*, come from the first account, while* 20% *come from the second, and the remaining* 10% *come from the third. Among the emails in account 1, only* 1% *are spam, while the spam rates for accounts 2 and 3 are* 2% *and* 5% *respectively. An email is chosen at random; what is the probability that it is spam?*

Bayes' Formula

Definition 2.3 (Bayes' Formula)

Let A_i $(i = 1, ..., n)$ be an exhaustive system of events, and B be an event such that $P(B) > 0$. Then for every i $(i = 1, ..., n)$:

$$P(A_i|B) = \frac{P(A_i)P(B|A_i)}{P(B)} = \frac{P(A_i)P(B|A_i)}{\sum_{i=1}^{n} P(A_i)P(B|A_i)}. \tag{10}$$

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Example 23

There are 10 lottery tickets, 4 of which are winning tickets. Student A draws first, and B draws second.

(a) *Is the game fair?*

(b) *If B wins, what is the probability that A also won?*

Example 24 (**Monty Hall problem**)

In a game show, a player must choose between 3 different doors, only one of which hides a prize. The host knows what is behind the doors. After the player picks a door, the host opens one of the remaining doors (not the one chosen by the player) that does not have a prize. The host then asks if the player wants to switch their choice. How should the player choose (stay or switch) to maximize the probability of winning the prize?

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Example 25

A student takes a multiple-choice history exam where each question has 5 options and only one correct answer. For any question, the student either knows the answer or does not; if they do not know, they guess. Suppose the probability that the student knows the correct answer is $p = 1/2$. Given that the student answered a question correctly, find the probability that they actually knew the answer (rather than guessing).

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Total Probability Rule. Bayes' Formula

Example 26

A person suspects they have disease B and goes to the hospital for a diagnostic test. It is known that the test is 95% accurate for those who actually have the disease. However, the false positive rate is 1% (i.e., if a healthy person is tested, there is a 0.01 probability the test says they are ill). Past epidemiological studies show that 5 out of every 1000 adults have disease B.

(a) *Calculate the probability that the person receives a positive test result.*

(b) *If the test result is positive, what is the probability that the person actually has the disease? What is the probability they are actually healthy?*